

NMMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Fifth Semester B.E. (E&E) (Credit System) Degree Examinations**

January 2023

20EE502 – SIGNAL ANALYSIS AND PROCESSING

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.**Unit – I****Marks BT* CO* PO***

1. a) Specify Nyquist rate for the following signals.
i) $x_1(t) = \sin(200\pi t)$ ii) $x_2(t) = \sin(300\pi t) + \cos(400\pi t)$ 4 L*2 1 1,2
- b) Sketch the following signals.
i) $y_1(t) = u(t) + u(t-1) - u(t-2) - u(t-3)$ ii) $y_2[n] = -u[n-1] + u[n-5]$ 6 L2 1 1,2
- c) Signal $x(t)$ is shown in Fig. 1c. Sketch the following signals.
i) $x_1(t) = x(t-4)$ ii) $x_2(t) = x(1-t)$ iii) $x_3(t) = x(\frac{1}{2}t+1)$

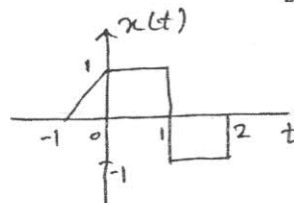


Fig. 1c

10 L3 1 1,2

2. a) Explain the sampling theorem and reconstruction of signals. 8 L2 1 1,2
- b) The trapezoidal pulse $x(t)$ shown in Fig. 2b is applied to a differentiator defined by $y(t) = \frac{d}{dt}x(t)$.
Draw the resulting output and find the total energy of $y(t)$.

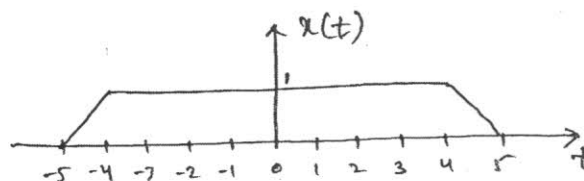


Fig. 2b

8 L3 1 1,2

- c) Consider the signals $x(t)$ & $y(t)$ shown in Fig. 2c. Sketch the following signal.

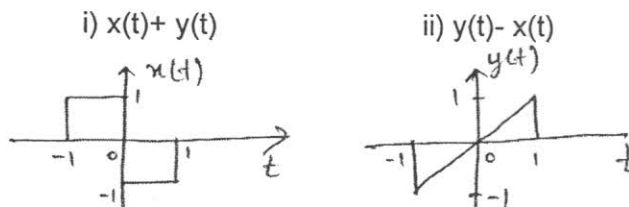


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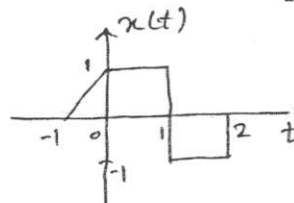


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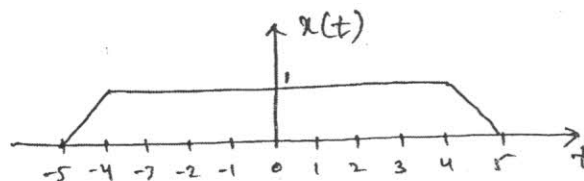


Fig. 2b

8 L3 1 1,2

- c) Consider the signals $x(t)$ & $y(t)$ shown in Fig. 2c. Sketch the following signal.

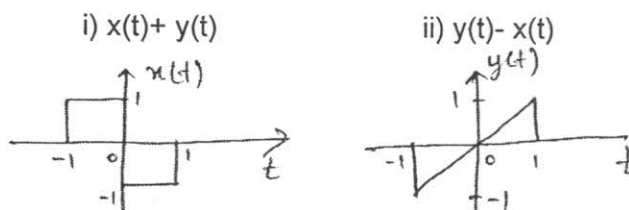


Fig. 2c

4 L2 1 1,2

Unit – IV

7. a) State and prove circular time shift property of DFT. 6 L2 4 1,2
- b) Let $X(k)$ be a 8-point DFT of a sequence $x[n]$. The first 4 samples of $X(k)$ are given by $X(0) = 3$, $X(1) = 5.8 - j2$, $X(2) = -7 + j10$, $X(3) = 0.2 + j2$, $X(4) = -9$. Find remaining samples. 4 L2 4 1,2
- c) Compute the circular convolution of the following sequence using DFT & IDFT method.
 $x_1[n] = \{1, 2, 3, 4\}$ & $x_2[n] = \{4, 3, 2, 1\}$. 10 L3 4 1,2
8. a) Using DIT-FFT algorithm compute DFT of the sequence.
 $x[n] = \{1, 1, 1, 1, 0, 0, 0, 0\}$ 10 L3 4 1,2
- b) Write a short note on
 i) Effect of Quantization in direct computation of DFT
 ii) Effect of Quantization on FFT algorithms. 10 L3 4 1,2

Unit – V

9. a) A filter is given by the difference equation
 $y[n] - \frac{1}{4}y[n-1] + \frac{1}{8}y[n-2] = x[n] + \frac{1}{2}x[n-2]$.
 Draw direct form-I & direct form-II realizations. Also obtain the transfer function of the filter. 10 L2 5 1,2
- b) Find the impulse response of a FIR filter with the following desired frequency response.
 $H_d(w) = \begin{cases} 0 & ; \text{if } |w| \leq \pi/6 \\ e^{-j4w} & ; \text{if } |w| > \pi/6 \end{cases}$
 Use rectangular window function. 10 L3 5 1,2, 3
10. a) Design an IIR digital Butterworth filter that when used will satisfy the following equivalent specification.
 i) low pass filter with -1dB cutoff 100π rad/sec
 ii) stop band attenuation of -35dB at 1000π rad/sec
 iii) Monotonic in stop band and pass band.
 iv) Sampling rate 2000 rad/sec
 v) Use bilinear transformation 10 L4 5 1,2, 3
- b) Differentiate between FIR and IIR filters. 4 L1 5 1,2
- c) Obtain linear phase realization of FIR filter described by difference equation
 $y[n] = x[n] - \frac{1}{2}x[n-1] + \frac{1}{4}x[n-2] + \frac{1}{4}x[n-3] - \frac{1}{2}x[n-4] + x[n-5]$. 6 L2 5 1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
